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Appendices

A DATASET DETAILS

A.1 TASKS DESIGN DETAILS

This subsection describes the task design details, aligning the original cognitive science psychometric test with the spatial abilities defined by Linn & Petersen (1985), and its classification within the Spatial-DISE taxonomy.

It is important to note that the Spatial-DISE taxonomy is intentionally a high-level, problem-centric abstraction. It categorizes spatial reasoning tasks based on their core structural properties and primary objectives—specifically, whether the problem relies on intrinsic versus extrinsic information, and whether the goal state requires static analysis or dynamic transformation.

However, human solvers and models may employ a variety of lower-level, atomic cognitive strategies to solve these tasks. For instance, while “3D Shape Finding” is structurally classified as an Intrinsic-Static task (as the objective is to determine a fixed intrinsic property of a single object), a viable solution strategy might involve mentally rotating the object (an Intrinsic-Dynamic process) to build a complete mental model. The DISE categorization reflects the dominant structural requirement of the task itself, whereas the classic spatial abilities (such as Mental Rotation [MR] or Spatial Visualization [SV] detailed in Table 5) represent the atomic skills that can be invoked across multiple quadrants. Therefore, our evaluation uses DISE to provide a coarse-grained organizational framework over task families, diagnosing systematic weaknesses along structural dimensions rather than isolating a single low-level cognitive process.

Table 6: Each spatial task used in our study and its canonical source test. Spatial Perception (SP), Spatial Relation (SR), Spatial Orientation (SO), Mental Rotation (MR), and Spatial Visualization (SV)

Task	Original Test	DISE Taxonomy	Spatial Ability
3D Combination	Differential Aptitude Tests (Bennett et al., 1947)	Extrinsic-Dynamic	SV
2D Combination	Minnesota Paper Form Board Test (Peter, 1974)	Extrinsic-Dynamic	SV
3D Projection	Purdue Spatial Visualization Test – Views (BODNER & GUAY, 1997)	Extrinsic-Static	SP, SV
Fold and Punch	Paper Folding Test (VZ-2) (Pallrand & Seeber, 1984; Ekstrom & Harman, 1976)	Intrinsic-Dynamic	SV, SR
3D Folding	Paper Folding Test (VZ-3) (Ekstrom & Harman, 1976)	Intrinsic-Dynamic	SV, SR, SO
2D Folding	Paper Folding Test (VZ-2) (Ekstrom & Harman, 1976)	Intrinsic-Dynamic	SV, SR, SO
3D Rotation	Mental Rotations Test (Shepard & Metzler, 1971)	Intrinsic-Dynamic	SV, MR, SO
2D Rotation	Card Rotations Test (S-1) (Ekstrom & Harman, 1976)	Intrinsic-Dynamic	SV, MR, SO
3D Shape Finding	Cube Comparisons Test (Ekstrom & Harman, 1976)	Intrinsic-Static	SV, SR
2D Shape Finding	Embedded Figures Test (Witkin, 1950)	Intrinsic-Static	SV, SR

A.2 DATASET SPLIT DETAILS

Subset	Q&A Pairs	Source Mix (RWD / SD)	Tasks
Spatial-DISE-Bench	559	53% / 47%	2D + 3D
Spatial-DISE-12K	12355	5% / 95%	3D
-Train	8648	5.1% / 94.9%	3D
-Val	1853	5.5% / 94.5%	3D
-Test	1854	4.5% / 95.5%	3D

Table 7: Description of Spatial-DISE Subsets. RWD: Real-World Data, SD: Synthetic Data. Note that Spatial-DISE Bench includes 2D questions absent from training splits, enabling zero-shot 2D evaluation.

A.3 REAL-WORLD DATA COLLECTION DETAILS

Real-world data are collected from open source online resources, mainly from the following resources:

1. Open-Source Spatial Reasoning Tests: Psychometric test materials published by academic research entities for evaluating spatial abilities.
2. CEM 11+ Non-verbal Reasoning Tests: Validated spatial reasoning items from authoritative aptitude tests used for secondary school admissions in the UK.
3. Online Employment Aptitude Tests: High-quality spatial and logical problems administered by corporations during recruitment.